Technical Report: Customer Churn Analysis

Capstone Project, Week 8 - end-to-end business analysis

1. Business problem and approach

The business question is: *where do we lose customers, how much monthly revenue is at risk, and what retention plan should the company implement?* The analysis follows five phases: planning, data preparation, exploratory analysis, advanced analysis (hypothesis tests, risk segmentation, prediction models) and recommendations. Success means identifying who churns, quantifying the revenue at risk, and proposing a plan whose value can be tested.

2. Data and quality checks

The file `customer_churn.csv` has 500 rows and 9 columns: CustomerID, Tenure (months), MonthlyCharges, TotalCharges, Contract, PaymentMethod, PaperlessBilling, SeniorCitizen and Churn (0 or 1). It meets the 500-row minimum.

Check	Result	Status
Rows	500	OK
Missing values / duplicate rows / duplicate IDs	0 / 0 / 0	OK
Churn values other than 0 or 1	0	OK
Tenure outside 0-72 months	0	OK
TotalCharges within 10% of Tenure x MonthlyCharges	6.4% of rows	PROBLEM

Data issue: TotalCharges should be close to Tenure x MonthlyCharges in real billing data, but only 6.4% of rows agree within 10%. The column was kept in the file, flagged, and **excluded from all analysis**. Cleaning consisted of trimming text fields and removing duplicates (none found). New columns: TenureGroup (0-11, 12-23, 24-47, 48+ months), FirstYear (tenure of 11 months or less), ChargeBand (quartiles) and Segment. Raw and cleaned files are saved in the `data/` folder.

3. Exploratory analysis

Overall churn is 10.6% (53 of 500). Every customer who left had a tenure of 11 months or less, so churn is a first-year problem.

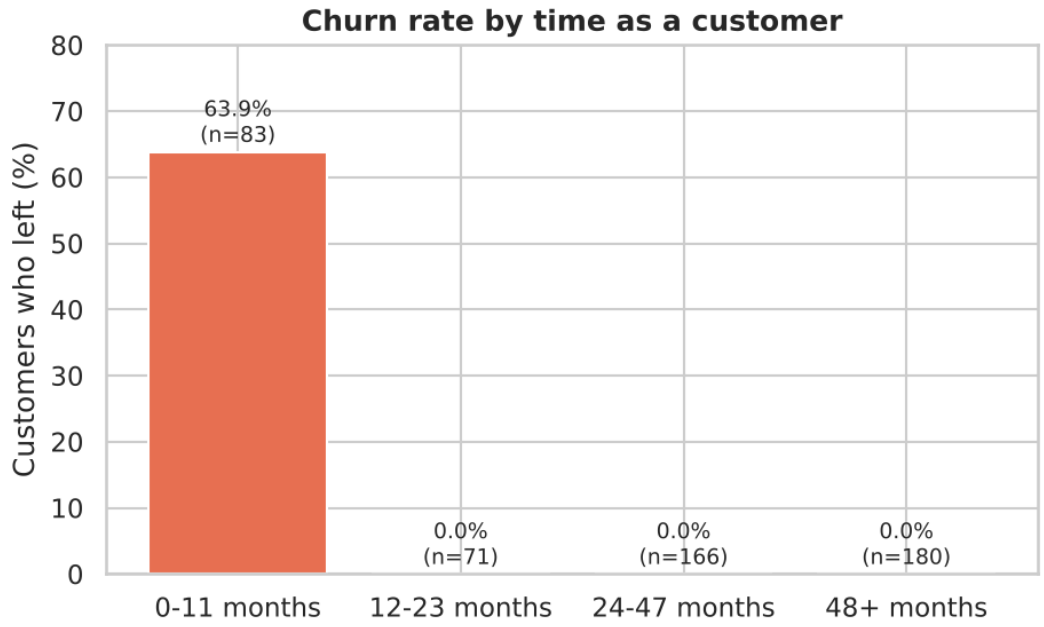


Figure 1. No customer beyond month 11 left. Churn among 0-11 month customers is 63.9% (53 of 83).

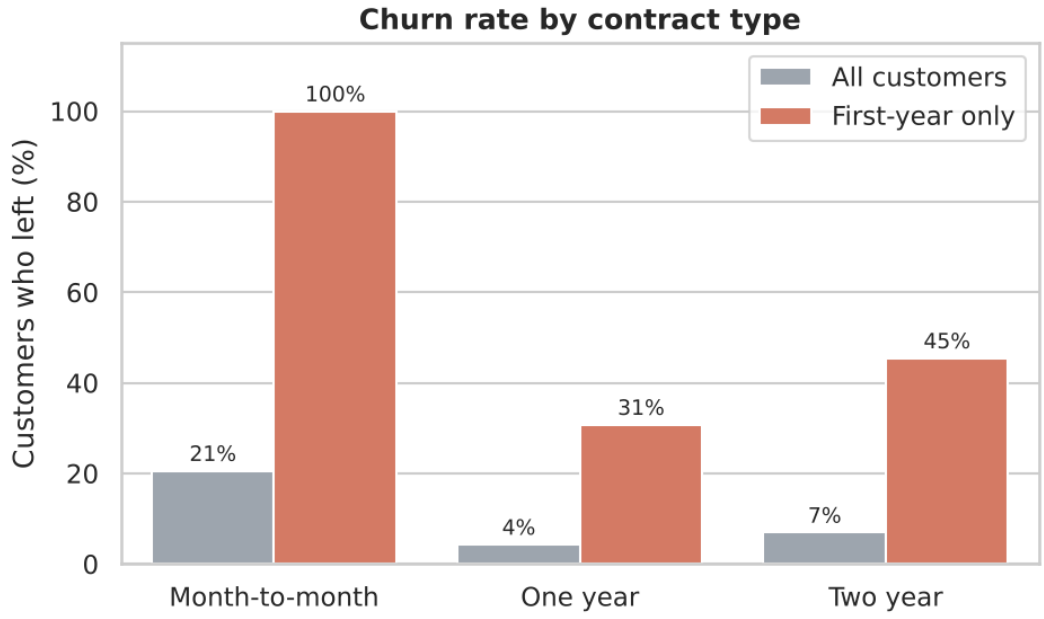


Figure 2. Month-to-month customers churn most overall (20.6%), but within the first year the 1-year and 2-year contract groups also lose customers.

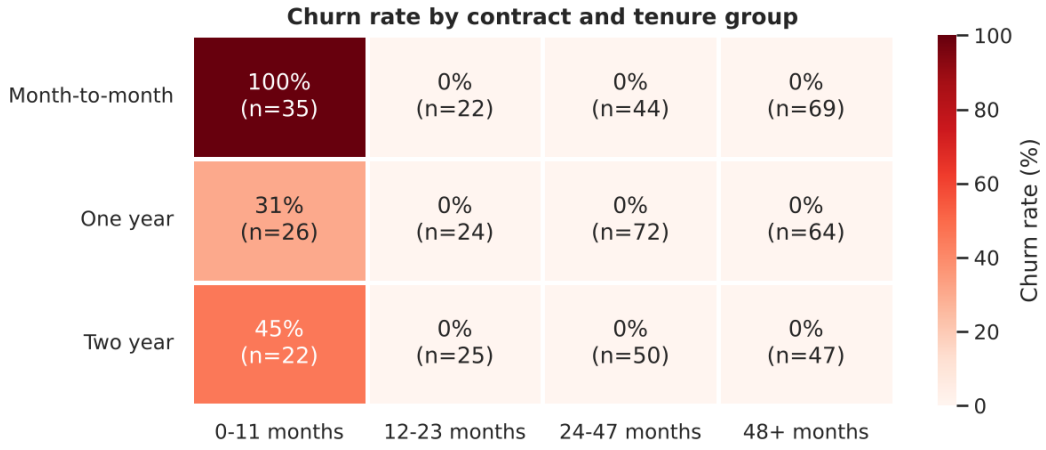


Figure 3. The risk is concentrated in the first-year column for every contract type.

Monthly charges also differ: churn is 6.3% in the lowest charge quartile, 9.5% in the second, 14.4% in the third and 12.2% in the highest. Payment method ($p = 0.14$), paperless billing ($p = 0.83$) and senior citizen status ($p = 0.80$) show no statistically clear link with churn (chi-square tests).

4. Advanced analysis

Technique 1: hypothesis tests (alpha = 0.05, Bonferroni correction for 3 tests).

Test	Result	p-value	Adjusted p
A. Churn depends on contract type (chi-square)	chi-square = 27.72	9.6e-07	2.9e-06 (survives)
B. First year: month-to-month vs other contracts (Fisher exact)	35 of 35 left vs 18 of 48	2.2e-10	6.7e-10 (survives)
C. Churners pay different monthly charges (Mann-Whitney)	median 137 vs 113	0.018	0.055 (does NOT survive)

Tests A and B are strong. Test C is significant on its own but does not survive the correction for running three tests, so the charge difference is treated as suggestive, not proven.

Technique 2: risk segmentation.

Segment	Customers	Left	Churn rate	Monthly revenue lost
Critical: first year + month-to-month	35	35	100%	4,233
High: first year + 1 or 2 year contract	48	18	37.5%	2,645
Stable: 12 months or more	417	0	0%	0

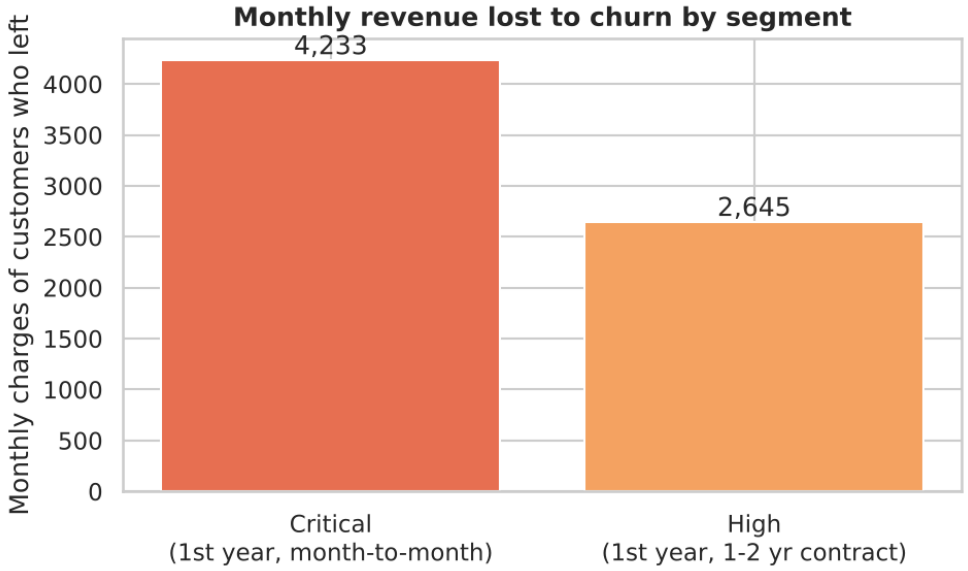


Figure 4. 61.5% of the lost monthly revenue (4,233 of 6,878) comes from the Critical segment, which is only 7% of customers.

Technique 3: prediction models. Logistic regression with 5-fold stratified cross-validation was compared on two feature sets. Model A (with Tenure) reaches ROC-AUC 0.98. Model B (only information known at sign-up: contract, charges, payment method, paperless billing, senior citizen) reaches 0.69. Model A is likely too optimistic: for customers who left, Tenure is the time they stayed before leaving, so it partly encodes the outcome, and the file does not say how it was recorded. Model B is the fairer estimate of what could be predicted for a new customer, and it is a modest signal. The segment view is therefore the more reliable basis for action than individual-level predictions.

5. Revenue at risk and retention scenarios

Customers who left represent 6,878 in monthly charges (12.1% of the 56,818 total), about 82,500 over 12 months. The scenarios below use **assumptions, not facts from the data**: programme cost of 0.5 (low) or 1.0 (high) month of average charges per enrolled customer, and a saved customer worth 12 months of charges.

Strategy	Share of leavers kept	Net benefit, low cost	Net benefit, high cost
Critical segment only	10%	2,963	847
Critical segment only	25%	10,582	8,466
Critical segment only	40%	18,202	16,085
All first-year customers	10%	3,428	-1,397
All first-year customers	25%	15,808	10,983
All first-year customers	40%	28,189	23,363

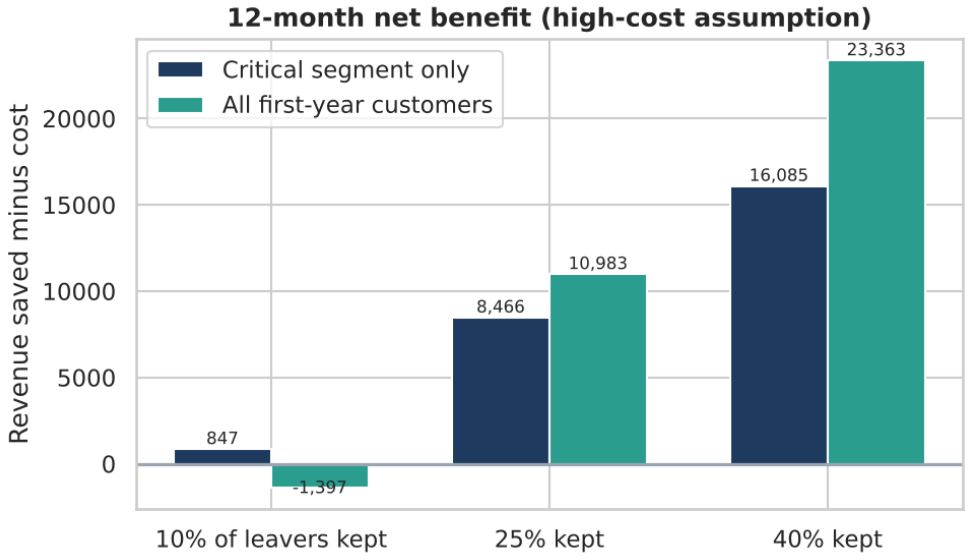


Figure 5. Net benefit is positive in 11 of 12 scenarios. The only loss is the costly offer to all first-year customers that keeps 10% of leavers.

Break-even: the Critical-segment programme pays for itself if it keeps 4% (low cost) to 8% (high cost) of would-be leavers; the all-first-year programme needs 6% to 12%.

6. Recommendations and implementation plan

- **Focus on the first 12 months.** Start with the Critical segment, which has the lowest break-even and the largest revenue loss per customer enrolled.
- **Run a 60-day pilot with a control group** (half of new customers enrolled), then measure first-year churn and cost per customer.
- **Scale only if the pilot pays for itself** (use the break-even figures above); otherwise change the offer and test again.
- **Fix data capture:** repair TotalCharges and record how Tenure is measured for customers who leave.

7. Limitations

- Only 500 customers and 53 churn events; the Critical segment has 35 customers, so rates like 100% are based on small numbers.

- Tenure may be recorded differently for customers who left, which could inflate how clearly churn separates by tenure.
- No data on programme cost or effect: all financial results are scenarios.
- TotalCharges is unreliable and was excluded. Currency is not stated in the file, so amounts are shown without a currency symbol.